AI-Enhanced Medicare Advantage Decision Support: A Multi-Agent Debate System with Local LLMs (gpt-oss-20b via Ollama)

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Abstract

Selecting a Medicare Advantage Prescription Drug (MAPD) plan is a complex, multi-factor decision that most beneficiaries make with incomplete information and tools that are not personalized. We present an AI-enhanced decision support system that combines (i) natural-language preference extraction, (ii) a multi-agent debate framework (Plan Advocate, Plan Challenger, Arbitrator), and (iii) county-level geographic intelligence (FIPS-driven coverage validation) to produce transparent, personalized plan recommendations. A key contribution is the system's *local* Enhanced AI analysis: we deploy the **gpt-oss-20b** large language model entirely *on-device* via **Ollama**. This eliminates reliance on third-party APIs (no OpenAI/Claude tokens), avoids per-call costs, and prevents any transmission of sensitive inputs off the user's machine, reducing security and privacy risk. We include an accessible web interface and demonstrate the approach with example scenarios.

1 Introduction

Medicare Advantage plan selection involves trade-offs across premiums, deductibles, prescription coverage, provider networks, wellness benefits, and service areas. Public tools (e.g., Medicare Plan Finder) provide extensive data but limited personalization and explanation. We seek to make plan choice more *intuitive*, *transparent*, and *tailored* by letting users express needs in plain English and by leveraging AI to turn those needs into defensible recommendations.

This paper documents a practical implementation of a Medicare Advantage decision support tool with:

- A modern web interface for location selection (state, county) and natural-language preferences,
- An NLP pipeline for extracting health conditions, budget constraints, lifestyle factors, and hard requirements,
- A two-mode evaluator: fast "Quick Analysis" scoring and a local "Enhanced AI" multi-agent debate using **gpt-oss-20b** via **Ollama**, and
- County-level (FIPS) geographic validation to ensure recommendations respect service areas.

2 System Overview

2.1 User Experience

The web app presents a simple form: pick state and county, then describe needs in natural language (e.g., "I'm on a fixed income, need strong diabetes coverage, and prefer a large cardiology network"). The system returns a ranked list of plans with explanations.

2.2 Core Architecture

Figure 1 shows a sample output screen. Internally, the system comprises the following modules:

- **Location Data & Validation** (`location_data.py`): state/county metadata, FIPS codes, service-area checks.
- **NLP Processor** (`nlp_processor.py`): entity extraction (conditions, drugs), constraint parsing, priority inference.
- **Dynamic Evaluator** (`dynamic_evaluator.py`): weights criteria based on extracted preferences; supports hard constraints.
- **Multi-Agent Debate** (`multi_agent.py`, `tournament.py`): Plan Advocate vs Challenger with an Arbitrator for close calls.
- **Web Interface** (`gradio_app.py`, `web_interface.py`): reactive UI with state/county dropdowns and rich HTML output.

3 Data and Geographic Intelligence

We use nationwide state/county metadata and *FIPS* codes to look up candidate plans, validate service areas, and tie recommendations to a user's actual location. This avoids suggesting ineligible plans and supports consistent cross-county comparisons.

4 Natural Language Preference Processing

We combine rule-based patterns and spaCy-based entity recognition to parse user descriptions into structured intent:

- **Health conditions** (e.g., diabetes, hypertension, CKD),
- **Budget priorities** (e.g., lowest premium, max out-of-pocket),
- **Lifestyle/choice** (e.g., large provider network, specific specialties), and
- **Hard constraints** (e.g., "$0 premium only").

The dynamic evaluator converts these into weights over plan attributes (cost, drug coverage, network breadth, benefits) and enforces hard constraints.

5 Plan Evaluation and Scoring

5.1 Two Analysis Modes

Quick Analysis provides fast, deterministic ranking using weighted attributes.
Enhanced AI Analysis (new) runs a *local* multi-agent debate powered by **gpt-oss-20b** under **Ollama**. Each plan is advocated and challenged; an Arbitrator provides a rationale for the winner. This yields richer explanations and can surface nuanced trade-offs.

5.2 Explainability

Outputs include per-plan reasoning: why a plan scored well (or poorly) with respect to the user's stated priorities, improving transparency beyond raw numeric scores.

6 Local LLM Deployment (Privacy, Cost, and Security)

A distinguishing feature is that Enhanced AI runs *entirely locally*. We deploy **gpt-oss-20b** via **Ollama**, so:

- **No external API usage:** *No* OpenAI or Claude calls; therefore **no token costs** and **no API keys** to manage.
- **Zero data egress:** User inputs (which may include sensitive health information) *never* leave the device, reducing risk and aiding HIPAA-aligned practices.
- **Deterministic fallback:** If the local model is unavailable, the system automatically falls back to Quick Analysis, which is also local.

This architecture delivers advanced reasoning while maintaining privacy-preserving, cost-free operation during analysis.

7 Web Interface and Example Output

The web UI is implemented with Gradio 4.x, providing responsive layout, state/county dropdowns with dynamic population, progress indicators, and accessible labels. A production system would most likely implement a different system such as Dash/Plotly within a cloud-based web app type of structure. A sample output screen is shown in Figure 1.

8 Example Scenarios

We illustrate three common personas (paraphrased from the README) and how the system tailors recommendations:

1. **Budget-Conscious Senior** (Miami, FL): "Cheapest plan that covers my blood pressure meds." The system emphasizes premiums and drug coverage; rankings reflect total annual cost.
2. **Active Retiree** (Denver, CO): "Very active, prefer strong cardiology network." Rankings prioritize network breadth and specialty access.

3. **Health-Focused User** (Phoenix, AZ): "Diabetes and kidney disease; comprehensive coverage." Emphasis shifts to formulary depth and specialist access, with cost secondary.

9 Implementation Notes

9.1 Installation and Startup

We provide both a simple startup and an integrated path that manages the local LLM:

- **Install (uv recommended):** `uv pip install -e .`
- **Basic Web:** `python gradio_app.py` (or installed entry point: `ma-web`)
- **Integrated Startup:** `python start_medicare_app.py` checks for Ollama, pulls `gpt-oss:20b` if needed, starts the server, and launches the web UI with Enhanced AI.

9.2 Key Dependencies

Gradio (web UI), spaCy (NLP), `us` (state metadata), `addfips` (county FIPS), Pydantic (models). All analysis is local; Enhanced AI uses `gpt-oss-20b` via Ollama.

10 Limitations and Future Work

This release uses representative sample data for demonstration (e.g., Denver County has the most complete dataset). Future work includes integration with CMS APIs, expanded formulary normalization, and cohort-based evaluation on real enrollment outcomes. We also plan A/B testing of Enhanced AI explanations versus Quick Analysis-only outputs for user comprehension and decision confidence.

11 Conclusion

We demonstrated a practical, privacy-preserving Medicare Advantage decision support system that combines natural language preference parsing, local LLM-based reasoning, and geographic intelligence to deliver transparent, personalized plan recommendations. Running **gpt-oss-20b** locally via **Ollama** eliminates external API costs and data-leak risk while enabling rich, explainable comparisons. The architecture is ready for production data sources and can generalize to adjacent payer and benefits selection domains.

Acknowledgments

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References

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🏥 Medicare Advantage Plan Finder

Find the best Medicare plan for your needs using AI-powered comparison

Active: 🤖🤖 Enhanced AI Analysis
Advanced multi-agent debate system using gpt-oss:20b

🔄 Refresh Models

Available Models:

⚡⚡ **Quick Analysis**
Fast plan comparison using weighted scoring • Instant results

🤖🤖 **Enhanced AI Analysis**
Advanced multi-agent debate system using gpt-oss:20b • Detailed reasoning

📍 Your Location

State
Select your state
Colorado (CO)

County in CO
Found 10 counties
Denver County (08031)

💬 Tell Us About Yourself

Maximum Monthly Premium ($)
What's the most you're willing to pay per month? Default is $0 (no premium).
100

Your Health & Preferences
Tell us about your health conditions, lifestyle, and what matters most to you

I am a 75 year old active adult. I run, exercise and play tennis. I have a few daily meds but nothing severe. I prefer a plan that is focused on wellness and not health problems

🔍 Find My Best Plan

📊 Analysis Results

Status

✅ Analysis complete! Compared 4 plans in Denver County, CO and found your best match.

🏆 Your Best Medicare Plan

🗒️ What We Understood About You

Age: 75 | Lifestyle: Active/fitness-focused

🥇 Denver PPO (CO)

Denver Medicare Plus

📍 Plan Type: PPO

💰 **Cost Summary**

Monthly Premium: **$38.67**

Annual Deductible: **$484**

Out-of-Pocket Maximum: **$3031**

💡 **Estimated Annual Cost: $1536**

📊 Analysis Summary

- Analyzed **4** Medicare Advantage plans
- Location: **Denver County, CO**
- Comparisons performed: **3**
- Tournament rounds: **2**
- Analysis Engine: 🤖 **Enhanced AI Analysis (gpt-oss-20b)**

🔍 Comparison Details

Round 1: Colorado HMO (CO) vs Denver PPO (CO)

Winner: Denver PPO (CO)

Scores: Plan A: 0.4, Plan B: 0.8

Reasoning: Okay, so I'm trying to help this 75-year-old beneficiary decide between the Colorado HMO and Denver PPO plans. Let me break down each plan and see what factors are important here. First, the beneficiary is 75 years old with general health conditions. They're active in their fitness routine but are looking for something cost-conscious and practical benefits. So, I need to look at both plans' features that align with these preferences. Looking at Plan A, Colorado HMO, it has a monthly pr

Round 2: Rocky HMO-POS (CO) vs Colorado HMO (CO)

Winner: Rocky HMO-POS (CO)

Scores: Plan A: 0.8, Plan B: 0.4

Reasoning: Okay, so I'm trying to help this 75-year-old beneficiary decide between two HMO plans. Let me break it down step by step. First, let's look at the details. The beneficiary is 75 years old with general health, active fitness level, and they're cost-conscious but value practical benefits. So, their priorities seem to be affordability and getting good coverage without too much hassle. Looking at Plan A: Rocky HMO-POS (CO). The monthly premium is $39.37, which seems pretty low for an 75-ye

Round 3: Denver PPO (CO) vs Rocky HMO-POS (CO)

Winner: Denver PPO (CO)

Scores: Plan A: 0.8, Plan B: 0.4

Reasoning: Okay, so I'm trying to help this 75-year-old beneficiary decide between two Medicare plans, Plan A and Plan B. Let me break down each plan first. Plan A is the Denver PPO with a monthly premium of $38.67, an annual deductible of $484, and a maximum out-of-pocket expense of $3031. It's part of the Denver Medicare Plus network. On the other hand, Plan B is Rocky HMO-POS with a higher premium of $39.37, a larger deductible of $521, and a higher max out-of-pocket of $4034. Rocky Mountain He

💡 Why This Plan Is Best for You

For a 75-year-old with your preferences, **Denver PPO (CO)** is the recommended choice. Key features: Low monthly premium ($38.67), Low deductible ($484).

⚠️ **Important:** This analysis uses sample data for demonstration purposes. For actual Medicare plan enrollment, please visit Medicare.gov or consult with a licensed insurance agent.